

Comparison of R/Python for XGBoost

1. The following is the result of my execution in Python and R with the stated dataset.

Method used	Dataset size	Testing-set predictive performance	Time taken for the model to be fit
XGBoost in Python via scikit-learn and 5-fold CV	100		
	1000		
	10000		
	100000		
	1000000		
	10000000		
XGBoost in R – direct use of xgboost() with simple cross-validation	100	0.2000000	0.32
	1000	0.1000000	0.40
	10000	0.0755000	0.34
	100000	0.0724500	0.74
	1000000	0.0702000	4.00
	10000000	0.0697245	62.50
XGBoost in R – via caret, with 5-fold CV simple cross-validation	100		
	1000		
	10000		
	100000		
	1000000		
	10000000		

2. While